

Question:

A mother brings her 3-year-old daughter to the pediatrician because she is concerned about her development. She states that her daughter seemed to regress in her motor development. Furthermore, she states she has been having brief episodes of uncontrollable shaking, which has been very distressing to the family. During the subsequent work-up, a muscle biopsy is obtained which demonstrates red ragged fibers and a presumptive diagnosis of a genetic disease is made. The mother states that she has another 6-year-old son who does not seem to be affected or have any similar symptoms. What genetic term explains this phenomenon? Answer Choices: A. Allelic heterogeneity B. Phenotypic heterogeneity C. Genetic heterogeneity D. Heteroplasmy

Reasoning + Answer:

<think>red ragged fibers can be found in both types of this disease - myopathy. Both the mother and her 6-year old son are affected but they seem to have different manifestations. This means they are phenotypically different. Hence this is phenotypic heterogeneity. </think> <answer> B . Phenotypic heterogeneity </answer>

Reward score

0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6